WEEK 2 PROJECT

DATA CLEANING & DATA-QUALITY REPORT

A complete beginner-friendly project using Python, Pandas and a real-world student performance dataset

Project	Clean a real dataset and write a data-quality report
Dataset	Students Performance in Exams (Kaggle)
Tools	Python 3, Jupyter Notebook / Google Colab, Pandas, Matplotlib
Final outputs	Clean CSV, quality report, charts and README
Suggested time	4–6 hours (including learning and testing)

Prepared as a practical submission guide. Replace the student details on the title page if your mentor requires them.

1. Project brief

Objective: Take a real dataset, inspect it, identify data-quality problems, clean the data using Python, and create a short report explaining what was checked and what changed.

One-sentence hand-in: I will submit a cleaned student-performance CSV file, a data-quality report showing missing values, duplicates, invalid values and outlier checks, visualisations, and a README explaining how to run the project.

The project uses the **Students Performance in Exams** dataset. It contains student background information and scores in mathematics, reading and writing. It is suitable for beginners because the columns are understandable and the dataset is small enough to work with on a normal laptop.

2. Dataset source and columns

Download the CSV from Kaggle: Students Performance in Exams – Kaggle. Sign in to Kaggle if required, download the dataset ZIP, extract it, and place **StudentsPerformance.csv** in the same folder as your notebook or Python script.

Column	Meaning	Expected values / checks
gender	Student gender	Text category; remove extra spaces
race/ethnicity	Group category	Text category; remove extra spaces
parental level of education	Parent/guardian education	Text category
lunch	Lunch type	Standard or free/reduced
test preparation course	Preparation course status	Completed or none
math score	Mathematics score	Whole number from 0 to 100
reading score	Reading score	Whole number from 0 to 100
writing score	Writing score	Whole number from 0 to 100

3. What you will check

• **Shape:** Number of rows and columns. This tells you the dataset size.

• **Missing values:** Empty cells in each column. The script fills numeric gaps with the median and categorical gaps with the mode.

• **Duplicate rows:** Repeated complete records. The script removes exact duplicate rows and records how many were removed.

• **Inconsistent text:** Leading/trailing spaces and inconsistent capitalisation in text fields. The script trims spaces and standardises category text.

• **Invalid scores:** Scores outside the valid 0–100 range. The script marks these as missing and then fills them using the median.

• **Outliers:** Unusually low/high values using the IQR method. Outliers are flagged in the report; they are not automatically deleted because a high or low score can be genuine.

Important: A data-cleaning decision must be explained. Do not delete unusual values only because they look different. Check whether they are impossible, incorrect, or simply rare.

4. Project workflow

Step	Action	Output
1	Download and load the CSV	Raw dataset loaded
2	Inspect rows, columns, types and sample records	Initial data profile
3	Check missing values and duplicates	Issue counts
4	Clean text and validate score ranges	Cleaned data
5	Handle missing values and save CSV	StudentsPerformance_Cleaned.csv
6	Create charts and quality report	PNG charts + report.txt
7	Test the files and write README	Ready-to-submit folder

5. Set up your project

Option A – Google Colab: Open Google Colab, create a new notebook, upload StudentsPerformance.csv using the Files panel, then paste and run the code in Section 6.

Option B – Laptop: Install Python 3 and run `pip install pandas matplotlib` in Terminal / Command Prompt. Save the code below as `clean_student_data.py` in the same folder as the CSV. Run `python clean_student_data.py`.

Folder structure after running:

```
Week2_Data_Cleaning/
■■■ StudentsPerformance.csv
■■■ clean_student_data.py
■■■ StudentsPerformance_Cleaned.csv
■■■ data_quality_report.txt
■■■ missing_values_before.png
■■■ score_distributions.png
■■■ README.md
```

6. Complete Python code

Copy the full code below into one notebook cell or save it as clean_student_data.py. It generates the cleaned CSV, two charts and a text report using the actual downloaded dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path

INPUT_FILE = "StudentsPerformance.csv"
OUTPUT_FILE = "StudentsPerformance_Cleaned.csv"

# 1) Load the original dataset
df = pd.read_csv(INPUT_FILE)
original_rows, original_cols = df.shape
original_columns = list(df.columns)

# 2) Standardise column names
df.columns = (
    df.columns.str.strip()
              .str.lower()
              .str.replace(r"[^a-z0-9]+", "_", regex=True)
              .str.strip("_")
)

# 3) Initial data-quality checks
missing_before = df.isna().sum()
duplicates_before = int(df.duplicated().sum())

# 4) Clean text columns
text_cols = df.select_dtypes(include="object").columns
for col in text_cols:
    df[col] = df[col].astype("string").str.strip()
    df[col] = df[col].replace("", pd.NA)

# Standardise category capitalisation (for example, MALE -> Male)
for col in text_cols:
```

```
    df[col] = df[col].str.replace(r"\s+", " ", regex=True).str.title()

# 5) Convert score columns to numeric and validate 0–100
score_cols = ["math_score", "reading_score", "writing_score"]
invalid_score_counts = {}
for col in score_cols:
    df[col] = pd.to_numeric(df[col], errors="coerce")
    invalid_mask = df[col].notna() & ~df[col].between(0, 100)
    invalid_score_counts[col] = int(invalid_mask.sum())
    df.loc[invalid_mask, col] = pd.NA

# 6) Remove exact duplicate rows
df = df.drop_duplicates().copy()
duplicates_removed = duplicates_before

# 7) Fill missing values
for col in df.columns:
    if pd.api.types.is_numeric_dtype(df[col]):
        if df[col].isna().any():
            df[col] = df[col].fillna(df[col].median())
    else:
        if df[col].isna().any():
            mode_values = df[col].mode(dropna=True)
            fill_value = mode_values.iloc[0] if not mode_values.empty else "Unknown"
            df[col] = df[col].fillna(fill_value)

# 8) Final checks
missing_after = df.isna().sum()
final_rows, final_cols = df.shape
df.to_csv(OUTPUT_FILE, index=False)

# 9) Chart 1: missing values before cleaning
plt.figure(figsize=(10, 5))
missing_before[missing_before > 0].sort_values(ascending=False).plot(
    kind="bar", title="Missing Values Before Cleaning"
)
plt.ylabel("Number of missing cells")
plt.xlabel("Column")
plt.tight_layout()
plt.savefig("missing_values_before.png", dpi=150)
plt.close()

# 10) Chart 2: score distributions after cleaning
available_scores = [c for c in score_cols if c in df.columns]
df[available_scores].hist(figsize=(10, 6), bins=15, edgecolor="black")
plt.suptitle("Score Distributions After Cleaning")
plt.tight_layout()
plt.savefig("score_distributions.png", dpi=150)
plt.close()

# 11) Create a data-quality report from the real run
with open("data_quality_report.txt", "w", encoding="utf-8") as f:
    f.write("WEEK 2 - DATA QUALITY REPORT\n")
    f.write("=" * 38 + "\n")
    f.write(f"Input file: {INPUT_FILE}\n")
    f.write(f"Original rows: {original_rows}\n")
    f.write(f"Original columns: {original_cols}\n")
    f.write(f"Rows after cleaning: {final_rows}\n")
    f.write(f"Columns after cleaning: {final_cols}\n")
    f.write(f"Duplicate rows found: {duplicates_before}\n")
    f.write(f"Duplicate rows removed: {duplicates_removed}\n\n")

    f.write("MISSING VALUES BEFORE CLEANING\n")
    f.write(missing_before.to_string() + "\n\n")
    f.write("INVALID SCORE VALUES FOUND\n")
    for col, count in invalid_score_counts.items():
        f.write(f"{col}: {count}\n")

    f.write("\nMISSING VALUES AFTER CLEANING\n")
    f.write(missing_after.to_string() + "\n\n")
    f.write("CLEANING ACTIONS\n")
    f.write("- Standardised column names.\n")
    f.write("- Removed extra spaces from text values.\n")
    f.write("- Standardised category capitalisation.\n")
    f.write("- Converted score columns to numeric values.\n")
    f.write("- Replaced scores outside 0-100 with missing values.\n")
    f.write("- Removed exact duplicate rows.\n")
    f.write("- Filled numeric missing values with the median.\n")
    f.write("- Filled categorical missing values with the mode.\n")
    f.write("- Saved the cleaned dataset and charts.\n\n")
    f.write("OUTLIER NOTE\n")
    f.write("Outliers were not deleted automatically. Review unusual scores\n")
    f.write("with subject-matter context before deciding whether to remove them.\n")

print("Cleaning completed successfully!")
print("Cleaned file:", OUTPUT_FILE)
print("Report: data_quality_report.txt")
print("Charts: missing_values_before.png, score_distributions.png")
```

```
print("\nFinal shape:", df.shape)
print("\nMissing values after cleaning:\n", missing_after)
```

7. Understanding the important code

• `df.shape` — Returns the number of rows and columns.

• `df.head()` — Displays the first five records so you can inspect the data.

• `df.isna().sum()` — Counts missing cells column by column.

• `df.duplicated().sum()` — Counts exact duplicate rows.

• `drop_duplicates()` — Removes repeated complete rows.

• `median()` — Finds the middle numeric value; it is less affected by extreme values than the mean.

• `mode()` — Finds the most frequent category and is useful for filling text fields.

• `between(0, 100)` — Checks whether a score is within the allowed range.

8. Data-quality report – what to submit

The script creates data_quality_report.txt automatically. Open it after running the script and use its real counts in your final submission. Do not copy made-up numbers from an example.

Report interpretation:

• Dataset overview: state the actual row count, column count and file name.

• Missing values: identify which columns had missing values and how many.

• Duplicates: state how many duplicate records were found and removed.

• Validity: state how many invalid score values were detected in each score column.

• Cleaning methods: explain median imputation for numeric fields and mode imputation for categorical fields.

• Final quality: confirm the number of missing cells after cleaning and the final dataset dimensions.

• Limitations: explain that the IQR/outlier review is not an automatic deletion step and that the dataset may not represent every school or student population.

9. Charts and what they tell the reader

Chart 1 – Missing Values Before Cleaning: Shows which columns need attention. If the chart is empty, there were no missing values in the original file.

Chart 2 – Score Distributions After Cleaning: Shows how mathematics, reading and writing scores are distributed after cleaning. Look for the common score ranges and unusually low/high observations.

A chart should answer a question. These charts answer: “Where were missing values located?” and “How are the cleaned test scores distributed?”

10. Testing checklist

■ CSV file loads without a FileNotFoundError.

■ The printed original row and column counts are visible.

■ The report lists missing values and duplicates before cleaning.

■ The cleaned CSV file is created.

■ The report says whether missing values remain after cleaning.

■ Both PNG chart files are created and open correctly.

■ The cleaned CSV has the same expected column names and no score below 0 or above 100.

■ The script can be run again without manually editing the code.

11. README.md – copy this into your project

````
# Week 2: Data Cleaning

## Project
Cleaned the Students Performance in Exams dataset and generated a data-quality report.

## Tools
- Python
- Pandas
- Matplotlib
- Google Colab or Jupyter Notebook

## Dataset
Students Performance in Exams (Kaggle):
https://www.kaggle.com/datasets/spscientist/students-performance-in-exams

Download `StudentsPerformance.csv` and place it in this project folder.

## Run
Install dependencies:
```bash
pip install pandas matplotlib
```
Run:
```bash
python clean_student_data.py
```

## Outputs
- `StudentsPerformance_Cleaned.csv` – cleaned dataset
- `data_quality_report.txt` – actual data-quality counts and cleaning actions
- `missing_values_before.png` – missing-value chart
- `score_distributions.png` – score distribution charts

## Cleaning performed
- Standardised column names and text values
- Checked missing values and exact duplicate rows
- Validated score values to the 0–100 range
- Filled numeric missing values with the median
- Filled categorical missing values with the mode
- Flagged outliers for review rather than deleting them automatically

## Learning
I learned how to inspect a dataset, identify common quality issues, clean records using Pandas, create charts, and document the cleaning
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12. Final submission note

Use this short note when submitting your work (edit it after you run the project):

> “I completed the Week 2 Data Cleaning project using the Students Performance in Exams dataset. I inspected the dataset, checked missing values and duplicate records, standardised text fields, validated score ranges, handled missing values, and exported a cleaned CSV. I also generated a data-quality report and visualisations. Through this project, I learned how to use Pandas for practical data cleaning and how to explain data-quality decisions.”

Attach these items: the cleaned CSV, the generated text report, both chart images, the Python notebook/script, and README.md. If the portal accepts only one file, submit this PDF together with a ZIP containing the output files, if allowed.

13. Learning links

- Kaggle Learn – Pandas
- Pandas user guide
- Matplotlib tutorials
- Google Colab

- Markdown README guide

End of Week 2 project guide.